COLLEGE STUDENT
PERFORMANCE ANALYTICS

End-to-End Data Analytics Project Report

Python • SQL Server • Power BI

Prepared using educational ERP data across six engineering departments

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1. Project Overview

This project analyzes student academic performance, skill development, internship participation, placement readiness, and academic risk using educational ERP data from six engineering departments.

The objective is to transform raw student data into meaningful insights that support academic planning, placement initiatives, and institutional decision-making.

2. Dataset Summary

Metric	Value
Total Rows	1,460
Final Records (after cleaning)	1,403
Total Columns	45

Departments Included

- Computer Engineering
- Information Technology
- ENTC
- Civil Engineering
- Mechanical Engineering
- Electrical Engineering

Key Features

Academic

- CGPA
- SGPA
- Attendance
- Backlogs
- Internal Marks

Skills

- Certifications
- Projects
- Hackathons

- Internships

Placement

- Placement Status
- Package
- Company
- Resume Score
- Aptitude Score
- Mock Interview Score

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python.

Data Loading

Imported dataset using pandas.

Initial Exploration

Used:

```
df.info()

df.describe()

df.shape()

df.head()
```

[4]:

	Student ID	Admission Year	Academic Year	Department	Gender	Age	Category	City	District	Hostel Status	...	Placement Status	Placement Eligibility Status	Number of Placement Offers	Placement Role
0	CS001	2022	BE	Computer	Female	20	ST	Nagpur	Nagpur	Day Scholar	...	Placed	Eligible	1	Software Engineer
1	CE001	2024	SE	Civil	Female	18	ST	Pune	Pune	Hosteller	...	Not Placed	Not Eligible	0	Site Engineer
2	EC001	2023	TE	ENTC	Male	21	EWS	Mumbai	Mumbai	Day Scholar	...	Not Placed	Not Eligible	0	Embedded Engineer
3	CS002	2024	SE	Computer	Male	19	OPEN	Aurangabad	Aurangabad	Hosteller	...	Not Placed	Not Eligible	0	QA Engineer
4	IT001	2023	TE	IT	Male	19	OBC	Mumbai	Mumbai	Hosteller	...	Not Placed	Not Eligible	0	Software Engineer

5 rows × 45 columns

Placement Company	Placement Package LPA	Placement Month	Resume Score	Aptitude Test Score	Mock Interview Score
Persistent	4.41	Sep	83	83	62
IRCON	0.00	Aug	68	42	84
Bosch	0.00	Sep	71	50	87
Capgemini	0.00	Sep	74	100	89
TCS	0.00	Sep	47	51	55

Missing Value Handling

Checked and handled missing values using:

`fillna()`

`mean()`

`median()`

`mode()`

```
internship_status                0
internship_domain              747
internship_duration_months       0
internship_company             747
internship_stipend               0
placement_status                 0
placement_eligibility_status     0
number_of_placement_offers       0
placement_role                 895
placement_company              895
placement_package_lpa            0
placement_month                895
resume_score                     0
aptitude_test_score              0
mock_interview_score             0
dtype: int64
```

```
[12]: df["internship_domain"] = df["internship_domain"].fillna("N/A")
```

```
[13]: df["internship_company"] = df["internship_company"].fillna("N/A")
```

```
[14]: df.loc[df['admission_type'] != 'CAP','cap_round'] = df.loc[df['admission_type'] != 'CAP','cap_round'].fillna('N/A')
```

```
[15]: placement_cols = [
    'placement_role',
    'placement_company',
    'placement_month'
]

df[placement_cols] = df[placement_cols].fillna('N/A')
```

Duplicate Removal

Used Student_ID as the business key.

```
drop_duplicates()
```

```
[18]: print(df.shape)
      print(df["student_id"].nunique())
      print(df["student_id"].duplicated().sum())

(1460, 45)
1460
0
```

7. Duplicate Columns Removing

```
[19]: (df["city"]==df["district"]).all()

[19]: True

[20]: df.drop("district",axis=1, inplace=True)

[21]: df.shape

[21]: (1460, 44)
```

Column Standardization

Converted columns into snake_case.

Example:

```
[9]: df.columns= (df.columns.str.strip().str.lower().str.replace(" ","_"))

[10]: df.columns

[10]: Index(['student_id', 'admission_year', 'academic_year', 'department', 'gender',
             'age', 'category', 'city', 'district', 'hostel_status',
             'admission_type', 'cap_round', 'cet_score', 'jee_percentile',
             'ssc_percentage', 'hsc_percentage', 'diploma_background',
             'current_cgpa', 'current_sgpa', 'previous_sgpa', 'semester_number',
             'attendance_percentage', 'active_backlogs', 'total_backlogs',
             'internal_marks', 'certification_count', 'certification_domain',
             'hackathon_count', 'project_count', 'project_domain',
             'internship_status', 'internship_domain', 'internship_duration_months',
             'internship_company', 'internship_stipend', 'placement_status',
             'placement_eligibility_status', 'number_of_placement_offers',
             'placement_role', 'placement_company', 'placement_package_lpa',
             'placement_month', 'resume_score', 'aptitude_test_score',
             'mock_interview_score'],
            dtype='object')
```

Feature Engineering

Created:

- Academic_Status
- Attendance_Category
- Skill_Level
- Risk_Level

[26]:
```
# Academic Status
#creating Feature mapping operation on sgpa
def academic_status(row):

    if row["current_cgpa"] >= 8 and row["active_backlogs"] == 0:
        return "Excellent"

    elif row["current_cgpa"] >= 7:
        return "Good"

    elif row["current_cgpa"] >= 6:
        return "Average"

    else:
        return "At Risk"

df["academic_status"] = df.apply(academic_status, axis=1)
```

[27]:
```
# Attendance Category
#Based on the % of attendance mapping it in classes

def attendance_category(x):

    if x >= 90:
        return "Excellent"

    elif x >= 75:
        return "Moderate"

    else:
        return "Low"

df["attendance_category"] = df["attendance_percentage"].apply(attendance_category)
```

[28]:
```
# Skill Level
# mapping skill level based on the certification count of student
def skill_level(x):

    if x == 0:
        return "Beginner"

    elif x <= 2:
        return "Intermediate"

    else:
        return "Advanced"

df["skill_level"] = df["certification_count"].apply(skill_level)
```

4. Data Analysis using SQL

We performed analytical queries in SQL Server to answer key business questions.

1. Department-wise Average CGPA

Evaluated academic performance across departments.

```
select department ,
round(avg(current_cgpa),2) as average_cgpa from student
group by department
order by average_cgpa desc;
```

	department	average_cgpa
1	IT	7.97
2	ENTC	7.93
3	Computer	7.91
4	Civil	7.89
5	Mechanical	7.84
6	Electrical	7.82

2. Placement Rate by Department

Analyzed placement trends among departments.

```
SELECT
    department,
    COUNT(*) AS total_students,
    COUNT(CASE WHEN placement_status='Placed' THEN 1 END) AS placed_students,
    ROUND(COUNT(CASE WHEN placement_status='Placed' THEN 1 END)*100.0/COUNT(*),2) AS placement_r
FROM student
GROUP BY department
ORDER BY placement_rate DESC;
```

Results | Messages

	department	total_students	placed_students	placement_rate
1	IT	234	36	15.380000000000
2	Computer	235	33	14.040000000000
3	Electrical	234	29	12.390000000000
4	Civil	239	26	10.880000000000
5	ENTC	231	24	10.390000000000
6	Mechanical	230	18	7.830000000000

3. Internship Participation Analysis

Studied internship engagement patterns.

```
select internship_domain, count(*) as tot_int from student
where internship_status='Yes'
group by internship_domain
order by tot_int desc;
```

Results | Messages

	internship_domain	tot_int
1	N/A	251
2	Planning Intern	28
3	QA	25
4	Web Dev	23
5	Software Dev	20
6	VLSI Intern	20
7	Power Intern	20
8	Automation Intern	19
9	Embedded Intern	18
10	Data Analyst	17
11	Design Intern	17
12	Site Intern	17
13	Production Intern	13

4. Students with Backlogs

Identified academically vulnerable students.

```
select student_id,
       academic_year,
       department,
       current_cgpa,
       active_backlogs from student
where active_backlogs >0
order by active_backlogs desc;
```

	student_id	department	current_cgpa	attendance_percentage	active_backlogs
1	EE062	Electrical	8.91	86.48	5
2	ME062	Mechanical	6.03	76.25	4
3	CS077	Computer	6.71	85.5	4
4	IT005	IT	6.83	81.45	4
5	EE057	Electrical	6.93	89.94	4
6	EC065	ENTC	7.79	82.68	4
7	IT099	IT	8.9	87.04	4
8	CS186	Computer	8.91	69.78	4
9	EC097	ENTC	9	85.75	4
10	CS102	Computer	9.7	69.89	4

5. Certification Analysis

Measured student skill participation.

```sql
SELECT
    department,
    COUNT(*) AS total_students,
    SUM(certification_count) AS total_certifications,
    ROUND(AVG(certification_count),2) AS average_certifications
FROM student
GROUP BY department
ORDER BY total_certifications DESC;
```

	department	total_students	total_certifications	average_certifications
1	Electrical	234	1062	4
2	Computer	235	961	4
3	IT	234	948	4
4	ENTC	231	925	4
5	Mechanical	230	921	4
6	Civil	239	902	3

6. Placement Eligibility Analysis

Evaluated placement readiness.

	department	total_students	eligible_students	not_eligible_students	eligibility_percentage
1	ENTC	231	113	118	48.920000000000
2	IT	234	112	122	47.860000000000
3	Computer	235	106	129	45.110000000000
4	Mechanical	230	103	127	44.780000000000
5	Civil	239	107	132	44.770000000000
6	Electrical	234	100	134	42.740000000000

7. Resume Score Analysis

Studied employability indicators.

```
SELECT
    department,
    COUNT(*) AS total_students,
    ROUND(AVG(resume_score),2) AS average_resume_score,
    MAX(resume_score) AS highest_resume_score,
    MIN(resume_score) AS lowest_resume_score
FROM student
GROUP BY department
ORDER BY average_resume_score DESC;
```

	department	total_students	average_resume_score	highest_resume_score	lowest_resume_score
1	Civil	239	70	95	45
2	ENTC	231	70	95	45
3	Mechanical	230	70	95	45
4	IT	234	69	95	45
5	Computer	235	69	95	45
6	Electrical	234	69	95	45

8. Academic Risk Analysis

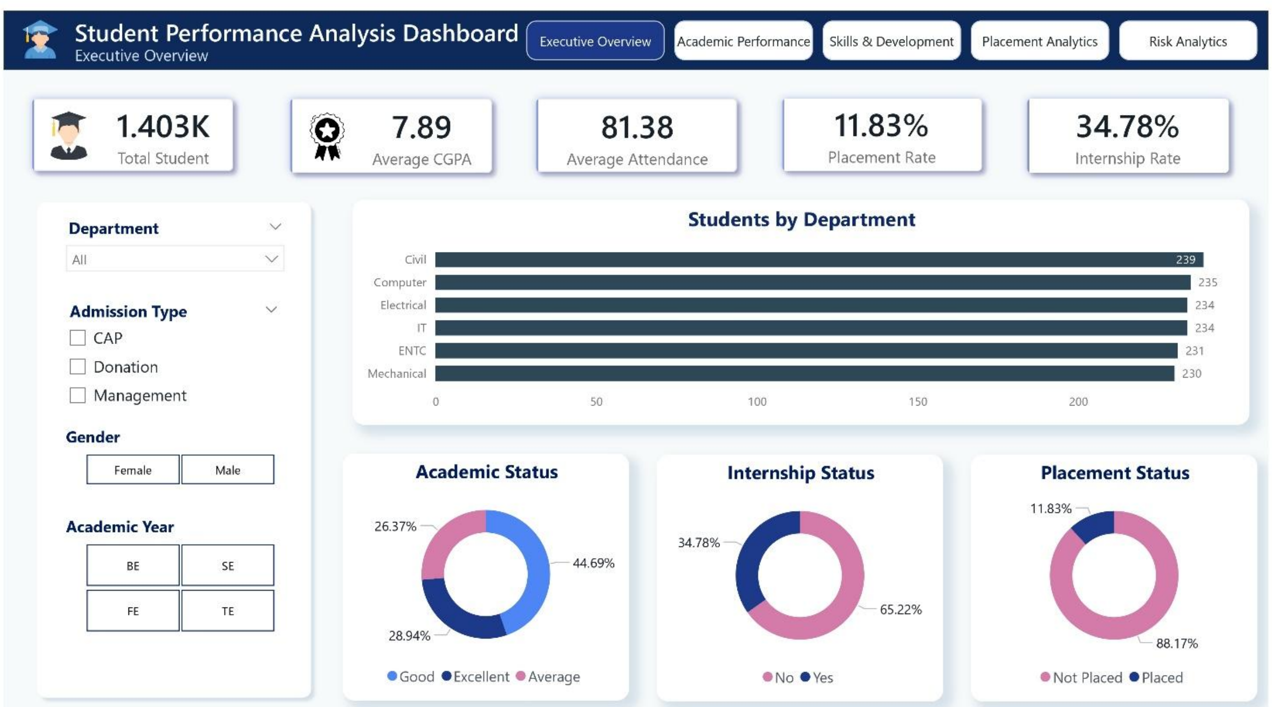
Identified students requiring intervention.

	student_id	department	current_cgpa	attendance_percentage	active_backlogs	risk_level
1	CS001	Computer	6.85	71.34	0	Medium Risk
2	CE001	Civil	6.84	80.4	0	Medium Risk
3	EC001	ENTC	7.06	73.88	0	Low Risk
4	CS002	Computer	8.29	93.21	0	Low Risk
5	IT001	IT	9.15	80.06	0	Low Risk
6	EC002	ENTC	9.74	80.13	2	Medium Risk
7	ME001	Mechanical	6.62	89.12	0	Medium Risk
8	ME002	Mechanical	7.76	87.03	0	Low Risk
9	EC003	ENTC	9.12	90.69	1	Medium Risk
10	IT002	IT	9.36	82.94	0	Low Risk

5. Dashboard in Power BI

Finally, we built interactive dashboards in Power BI to present insights visually.

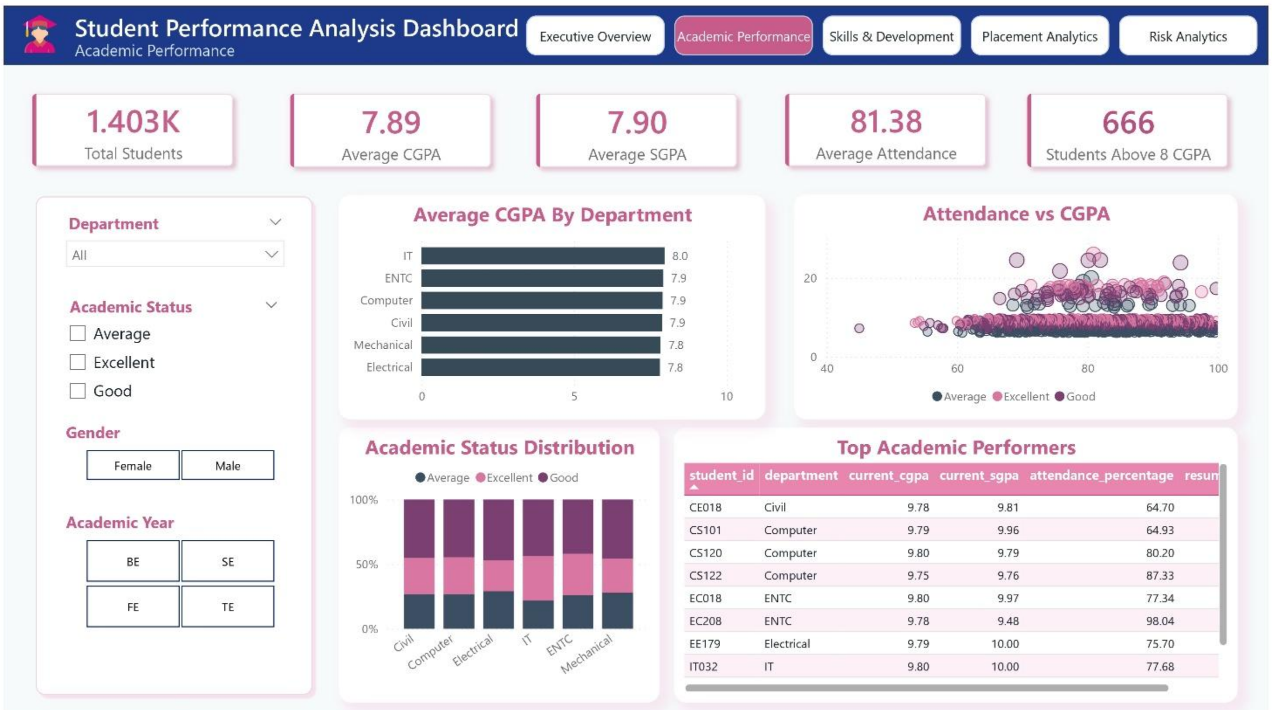
Executive Overview



KPIs

- Total Students
- Average CGPA
- Average Attendance
- Placement Rate
- Internship Rate

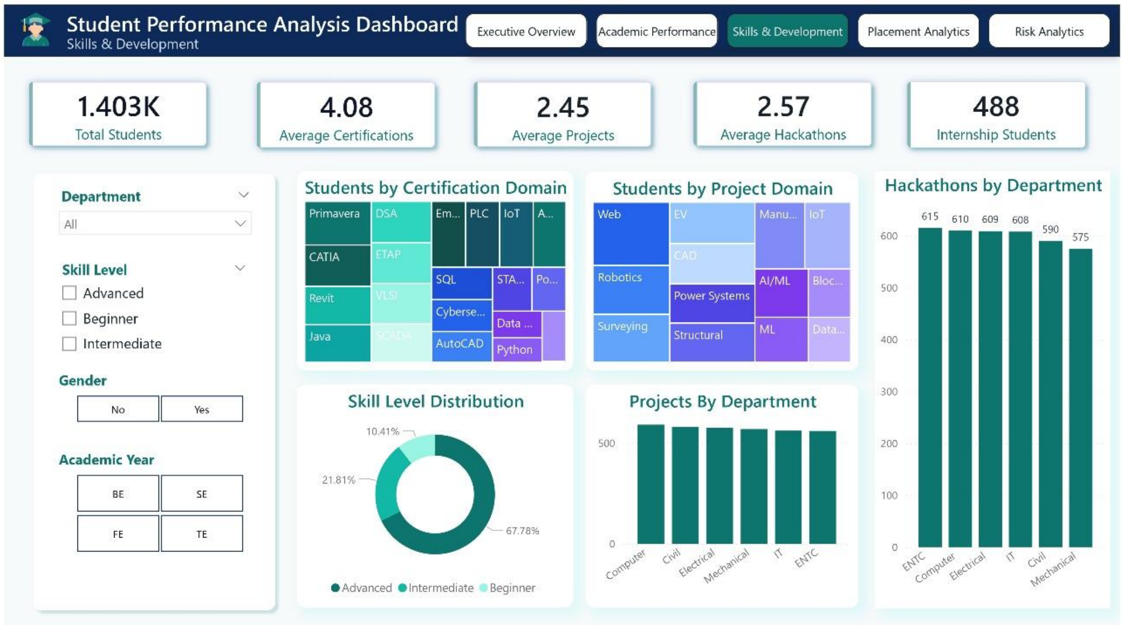
Academic Performance



KPIs

- Average CGPA
- Average SGPA
- Students Above 8 CGPA

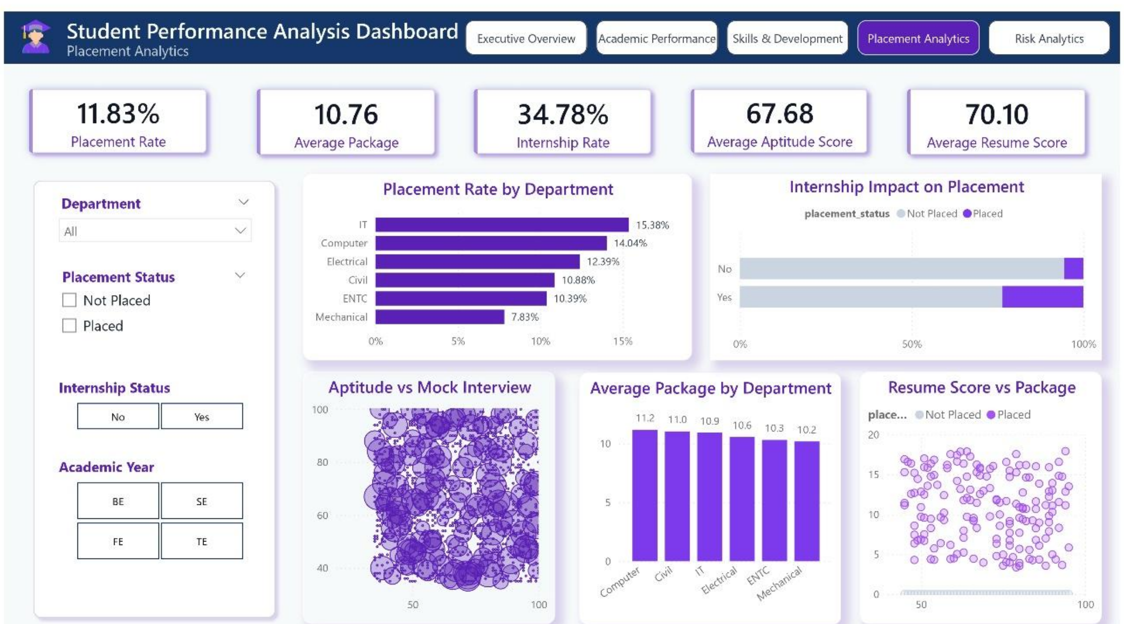
Skills & Development



KPIs

- Certifications
- Projects
- Hackathons
- Internship Students

Placement Analytics



KPIs

- Placement Rate
- Average Package
- Resume Score

Risk Analytics

KPIs

- High Risk Students
- Students with Backlogs
- Risk Distribution

6. Business Recommendations

- Improve attendance monitoring systems.
- Encourage certification participation.
- Expand internship opportunities.
- Conduct resume-building workshops.
- Implement early intervention mechanisms for at-risk students.
- Promote hackathons and project-based learning.

7. Conclusion

This project demonstrates an end-to-end educational analytics workflow using Python, SQL Server, and Power BI.

The developed dashboards provide actionable insights into academic performance, skill development, employability, and academic risk, supporting data-driven institutional decision-making.